



MATHEMATICS SSC-I (Science Group)

SECTION – A (Marks 15)

Time allowed: 20 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
1	0	0	8	1

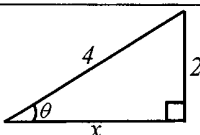
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ROLL NUMBER						

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6	6	6	6	6	6	6
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8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. $\left(\frac{1}{16}\right)^{-\frac{3}{2}} =$		$\frac{1}{4}$	4	16	64	☐	☐	☐	☐
2. 4500 in scientific notation:		4.5×10^3	45×10^3	4.5×10^2	0.45×10^5	☐	☐	☐	☐
3. Which one is a binary relation?		$\{1, 2, 3\}$	$\{(1, 2), (2, 3)\}$	$\{1 \rightarrow 2 \rightarrow 3\}$	$\{1, 2\} \times \{3\}$	☐	☐	☐	☐
4. Highest common factor of $-12x^2y^2$, $6xy^3$, $24x^2y^2$ is:		$-6xy$	$-6x^2y^3$	$6xy^2$	$24y^3x^2$	☐	☐	☐	☐
5. $x + 3 > 5$ implies:		$x = 0$	$x < 2$	$x \leq 2$	$x > 2$	☐	☐	☐	☐
6.  $\theta = ?$		90°	60°	30°	0°	☐	☐	☐	☐
7. If a plane flies on a bearing of 180° , it is moving towards:		North	South	East	West	☐	☐	☐	☐
8. If two triangles are equal, then their corresponding sides are:		Equal	Unequal	Unproportional	Perpendicular	☐	☐	☐	☐
9. $\frac{1}{\sin \theta} =$		$\tan \theta$	$\cos \theta$	$\sec \theta$	$\operatorname{cosec} \theta$	☐	☐	☐	☐
10. The points where the medians of a triangle intersect is called the:		Incentre	Circumcentre	Centroid	Orthocentre	☐	☐	☐	☐
11. A triangular road sign has all sides equal. What type of triangle is it?		Right-angled	Equilateral	Isosceles	Scalene	☐	☐	☐	☐
12. The locus of points equidistant from two given points is the:		Perpendicular bisector	Angle bisector	Median	Altitude	☐	☐	☐	☐
13. Class mark of interval (15 – 20) is:		15	17.5	20	35	☐	☐	☐	☐
14. The probability of an event CANNOT be:		0	0.5	1	2	☐	☐	☐	☐
15. A boy scored 3 goals in 9 matches. His relative frequency of scoring goals is:		9	3	$\frac{1}{2}$	$\frac{1}{3}$	☐	☐	☐	☐

—1SA-I 26008 (B) —



Mathematics SSC-I (Science Group)

62

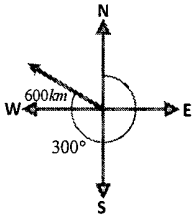
Time allowed: 2:40 Hours

Total Marks Sections B and C: 60

SECTION – B (Marks 36)

Q. 2 Attempt the following parts.

(9 x 4 =36)

(i)	Simplify the following expression using laws of indices: a. $2^{\frac{2}{3}} \cdot 2^{\frac{1}{7}}$ b. $3^{\frac{-1}{2}} \cdot 3^{\frac{1}{2}}$	2+2	OR	In a bag with 10 balls, there are 7 black and 3 white balls. If one ball is selected at random from the bag, calculate: a. The probability of selecting a white ball. b. The probability of selecting a black ball c. The sum of all the probability	04
(ii)	A plane travels 600 km on a bearing of 300° . How far west did the plane travel? 	04	OR	An earthquake that occurred in Pakistan in 2005, measured 7.6 on the Richter scale. In 1978, an earthquake in China measured 8.2 on the Richter scale. How many times, the China's earthquake was stronger than the Pakistan's earthquake? (Use $M = \log \left(\frac{I}{I_0} \right)$)	04
(iii)	If $A = \{1, 3\}$ and $B = \{2, 4\}$: a. Write the cartesian product $A \times B$. b. Find $R_1 = \{(x, y) x \in A, y \in B \wedge x > y\}$	3+1	OR	Find the equation of a line through the point (1,3) and the point of intersection of lines $2x - 3y + 4 = 0$ and $4x + y - 1 = 0$.	04
(iv)	Solve the inequality, and plot the solution on real number line. $-3x + 11 \geq 5(x + 7) - 32 \geq 3x - 1 \quad \forall x \in \mathbb{R}$	3+1	OR	Factorize the quadratic expression: $3x^2 + 16x - 12$	04
(v)	Solve the right-angled triangle $\triangle ABC$, $m\angle C = 90^\circ$, $a = 4\text{cm}$, $c = 4\sqrt{2}\text{cm}$	04	OR	Verify that $\sin^2 \theta + \cos^2 \theta = 1$	04
(vi)	Find the gradient of the line passing through the points $A(2, 3)$ and $B(6, 11)$.	04	OR	Prove by using Venn diagram $(P \cup Q) \cup R = P \cup (Q \cup R)$. $P = \{0, 1, 2, 3\}$, $Q = \{2, 3, 4, 5, 6\}$, $R = \{5, 6, 7, 8, 9\}$	2+2
(vii)	A water fountain is to be placed at a point which is equidistant from two lamp posts located at $P(4, 7)$ and $Q(10, 1)$. Find possible positions of the fountain.	2+2	OR	Find the equation of straight line passing through $(-4, -4)$ and parallel to the line with slope -5 .	04
(viii)	The ratio of areas of the bases of two similar cylinders is $9:4$. The area of base of the smaller cylinder is 240cm^2 . Calculate the radius, diameter and area of the base of large cylinder.	04	OR	The monthly incomes (in thousand rupees) of five workers are: 20, 22, 25, 90, 23. Calculate the mean and median. Which measure better represents the income level? Give reason.	1x4
(ix)	For algebraic expressions $P = 6x^2y$ and $Q = 9xy^2$ show that $P \times Q = \text{LCM} \times \text{HCF}$	1x4	OR	In a coin toss experiment, Head is obtained 15 times in 50 trials. Calculate the relative frequencies of Head and Tails.	2+2

SECTION – C (Marks 24)

Note: Attempt the following questions.

(3x8=24)

Q.3	Find the values of p and q if $x^4 - 12x^3 + px + q$ is a complete square.	08	OR	Find the interior angles of triangle whose vertices are $A(-2, 0)$, $B(3, 0)$ and $C(6, 5)$.	08												
Q.4	From the eye level of a 6 feet tall person standing on the ground, the angle of elevation of the top of a tower is 30° . a. What type of triangle is formed in this situation? b. Which trigonometric ratio would be used to find the height of the tower? c. Find height of tower from its base.	1+1 +6	OR	Convert $4x - 3y + 9 = 0$ in given forms: a. Slope intercept b. Two intercepts c. Symmetric d. Normal	2x4												
Q.5	Draw a triangle $\triangle ABC$ when $\overline{AB} = 5.8\text{cm}$, $\overline{BC} = 4.9\text{cm}$, $m\angle b = 60^\circ$ a. Draw angle bisectors of triangle. b. Are these angle bisectors concurrent? Verify.	3+4 +1	OR	Consider the data and find the mean. <table border="1" style="display: inline-table; margin-top: 5px;"> <tr> <td>CI</td> <td>140-150</td> <td>150-160</td> <td>160-170</td> <td>170-180</td> <td>180-190</td> </tr> <tr> <td>f</td> <td>6</td> <td>9</td> <td>13</td> <td>7</td> <td>5</td> </tr> </table>	CI	140-150	150-160	160-170	170-180	180-190	f	6	9	13	7	5	4+4
CI	140-150	150-160	160-170	170-180	180-190												
f	6	9	13	7	5												



MATHEMATICS SSC-I
(Science Group)
SECTION – A (Marks 15)
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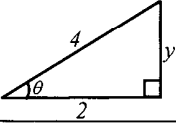
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ROLL NUMBER						

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9	9	9	9	9	9	9

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Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. $16^{\frac{1}{2}} =$	2	8	4	16	☐	☐	☐	☐
2. Correct scientific notation of 350 =	35×10^2	3.5×10^2	35×10^3	350×10^1	☐	☐	☐	☐
3. Domain of $\{(2,4), (3,5)\}$:	$\{2,3\}$	$\{4,5\}$	$\{3,4\}$	$\{2,4\}$	☐	☐	☐	☐
4. H.C.F of $8x^2$, $12x^3$ and $24x^2y^2$ is:	$4x^2$	$4x$	$8x^2$	$12x^3$	☐	☐	☐	☐
5. $x+5 > 9$ implies:	$x = 0$	$x > 4$	$x < 4$	$x = 4$	☐	☐	☐	☐
6.  $\theta =$	0°	30°	60°	90°	☐	☐	☐	☐
7. A ship moves from point A to B on a bearing of 90° . In which direction does it travel?	North	South	West	East	☐	☐	☐	☐
8. Two polygons are similar if they have:	Equal area	Equal perimeters	Equal corresponding angles	Equal volumes	☐	☐	☐	☐
9. Which of the following is a fundamental trigonometric identity?	$\sin \theta = \cos \theta$	$\sin^2 \theta + \cos^2 \theta = 1$	$\tan \theta = \sin \theta$	$\sec \theta = \cos \theta$	☐	☐	☐	☐
10. The angle bisectors of a triangle are concurrent at the:	Incentre	Orthocentre	Circumcentre	Centroid	☐	☐	☐	☐
11. The exterior angle of a regular polygon is 36° . number of sides in polygon are:	5	6	9	10	☐	☐	☐	☐
12. The locus of points at a fixed distance from a given point is a:	Straight line	Triangle	Circle	Altitude	☐	☐	☐	☐
13. The modal class in a grouped frequency distribution is the class with:	The highest frequency	The lowest frequency	The median frequency	The mean frequency	☐	☐	☐	☐
14. If an event has probability $\frac{3}{5}$, then the probability of the even not occurring is:	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{5}{3}$	$\frac{1}{5}$	☐	☐	☐	☐
15. In a class, 18 out of 40 students received an A grade. The relative frequency of grade A is:	$\frac{6}{40}$	$\frac{18}{40}$	$\frac{2}{40}$	$\frac{8}{40}$	☐	☐	☐	☐

—ISA-I 26008 (D) —



Mathematics SSC-I (Science Group)

Time allowed: 2:40 Hours

Total Marks Sections B and C:60

SECTION – B (Marks 36)

Q. 2 Attempt the following parts:

(9 x 4 =36)

(i)	Simplify: $\sqrt[4]{\frac{a^3}{b^3}} \times \sqrt[4]{\frac{b^3}{c^3}} \times \sqrt[4]{\frac{c^3}{a^3}}$	04	OR	$U = \{1, 2, 3, 4, 5, 6\}, A = \{2, 5, 6\}, B = \{1, 2, 3\}$, then use Venn diagrams to show that: $(A \cup B)' = A' \cap B'$	2+2
(ii)	A cyclist travels 10 km due south and then 24 km due west. What simple bearing should the cyclist take to return directly?	04	OR	In the figure, $\overline{BC} \parallel \overline{DE}$. find: a. Ratio $\frac{DE}{BC}$ b. Ratio $\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle ABC}$ c. Find the area of $\triangle ADE$ if area of $\triangle ABC$ is 256cm^2 . d. Area of trapezium $DBCE$.	1x4
(iii)	An earthquake of 1920 was measured about 8.5 on Richter scale and another earthquake of 1923 was measured 7.8 on that scale how many times stronger was the 1920 earthquake than 1923 earthquake? $(M = \log \left(\frac{I}{I_0} \right))$	04	OR	In a bag with 10 balls, there are 7 black and 3 white balls. If one ball is selected at random from the bag, calculate: a. The probability of selecting a white ball b. The probability of selecting a black ball c. The sum of all the probabilities	04
(iv)	Solve the inequality, and plot the solution on real number line. $3x + 5 \leq 5 - 3(x + 2) \leq 6x - 10 \quad \forall x \in \mathbb{R}$	3+1	OR	Factorize: $(x^2 + 5x + 4)(x^2 + 5x + 6) - 3$	04
(v)	Verify that: $\frac{\sqrt{1 - \cos \theta}}{1 + \cos \theta} = \frac{\sin \theta}{1 + \cos \theta}$	2+2	OR	An exterior angle of a regular polygon is 12° . What is the sum of all the interior angles? If length of its one side is 1 m then find its perimeter.	3+1
(vi)	Lines AB and CD with coordinates $A(0,3), B(2,8), C(3,10), D(8,k)$ are perpendicular. Find k .	04	OR	Find equation of line passing through point of intersection of lines. $3x + 2y + 1 = 0, x - 2y + 3 = 0$ and passing through point $(-1,0)$.	04
(vii)	A company wants to install a new cell phone tower so that it is equidistant from two existing towers located at $P(2,5)$ and $Q(8,3)$. Find equation of the locus where the new tower should be placed.	04	OR	Find the equation of the line that crosses the line $y = 2x - 5$ at right angle at the point $(3,1)$.	04
(viii)	If $A = \{1,3\}$ and $B = \{2,4\}$: a. Write the cartesian product $A \times B$. b. Find $R_1 = \{(x,y) x \in A, y \in B \wedge x > y\}$	2+2	OR	A fair coin is tossed 150 times and head comes up 70 times. What is the relative frequency of getting Heads. Also find the relative frequency of getting Tails.	2+2
(ix)	If $A = 4a^2b$ and $B = 6ab^2$ then show that $A \times B = \text{LCM} \times \text{HCF}$	2+2	OR	The number of items sold by a shop on different days are: 6, 8, 6, 5, 7, 6, 9. a. Find mode b. Find median c. Find mean	1+1 +2

SECTION – C (Marks 24)

Note: Attempt the following questions:

(3x8=24)

Q.3	Construct $\triangle ABC$ when $\overline{AB} = 5.8\text{cm}, \overline{BC} = 5\text{cm}, m\angle B = 45^\circ$. Draw its medians. Are these medians concurrent? Verify.	3+3 +1 +1	OR	Find the interior angles of triangle XYZ whose vertices are: $X(-3,2), Y(0,-1), Z(3,3)$	08												
Q.4	From the top of a tower of height 120 m, angles of depression of two boats on same side of tower at water level are 60° and 45° . Find distance between the boats.	08	OR	Convert the following equation into different standard forms: $x - 2y + 4 = 0$ a. Slope intercept b. Two intercepts c. Symmetric d. Normal	2x4												
Q.5	For what value of k , the expression $y^4 + 4y^2 + k + \frac{8}{y^2} + \frac{4}{y^4}$ becomes a perfect square, where $y \neq 0$.	08	OR	The given table shows distribution of marks obtained by 50 students in a mathematics test: Calculate the median marks.	08												
				<table border="1"> <tr> <td>CI</td><td>01-10</td><td>11-20</td><td>21-30</td><td>31-40</td><td>41-50</td></tr> <tr> <td>f</td><td>5</td><td>10</td><td>15</td><td>12</td><td>8</td></tr> </table>	CI	01-10	11-20	21-30	31-40	41-50	f	5	10	15	12	8	
CI	01-10	11-20	21-30	31-40	41-50												
f	5	10	15	12	8												



MATHEMATICS SSC-I (Science Group)

SECTION – A (Marks 15)

Time allowed: 20 Minutes

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Answer Sheet No. _____

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1	9	0	8	1

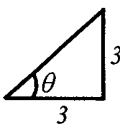
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9	●	9	9	9

ROLL NUMBER						

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8	8	8	8	8	8	8
9	9	9	9	9	9	9

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Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. $\left(\frac{1}{9}\right)^{-\frac{1}{2}}$ equals:		3	6	9	81	☐	☐	☐	☐
2. Scientific notation of 71000 is:		7.1×10^2	0.71×10^2	7.1×10^3	7.1×10^4	☐	☐	☐	☐
3. Range of $\{(1, 3), (2, 5)\}$ is:		$\{1, 2\}$	$\{3, 5\}$	$\{1, 3\}$	$\{2, 5\}$	☐	☐	☐	☐
4. For two polynomials A and B, $HCF \times LCM =$		$A + B$	$A - B$	$(HCF)^2$	$A \times B$	☐	☐	☐	☐
5. $-2x \leq 4$ gives:		$x \leq -2$	$x \geq -2$	$x \leq -1$	$x = -1$	☐	☐	☐	☐
6.  $\theta = ?$		45°	60°	90°	120°	☐	☐	☐	☐
7. A cyclist travels from P to Q on a bearing of 45° . This means the cyclist moves.		South-East	North-East	South-West	North-West	☐	☐	☐	☐
8. If two triangles are equal, then their corresponding sides are:		Equal	Unequal	Unproportional	Perpendicular	☐	☐	☐	☐
9. $1 + \tan^2 \theta =$		$\sin^2 \theta$	$\cos^2 \theta$	$\sec^2 \theta$	$\operatorname{cosec}^2 \theta$	☐	☐	☐	☐
10. The perpendicular bisectors of the sides of a triangle meet at the:		Incentre	Centroid	Orthocentre	Circumcentre	☐	☐	☐	☐
11. Each interior angle of a regular square-shaped garden is:		90°	60°	45°	30°	☐	☐	☐	☐
12. The locus of points equidistant from two intersecting straight lines is formed by the:		Two parallel lines	Perpendicular bisectors	Circles	Angle bisectors	☐	☐	☐	☐
13. The median of 17, 27, 37, 47, 57, 67, 77 is:		47	17	77	0	☐	☐	☐	☐
14. If a fair die is rolled once, what is the probability of getting 2?		$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{5}$	☐	☐	☐	☐
15. Which is a possible Valid relative frequency value:		-0.5	1.2	0.75	2	☐	☐	☐	☐

—1SA-I 26008 (M)—



Mathematics SSC-I (Science Group)

66

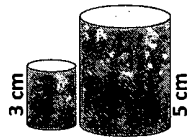
Time allowed: 2:40 Hours

Total Marks Sections B and C:60

SECTION – B (Marks 36)

(9 x 4 =36)

Q. 2 Attempt the following parts.

(i)	Simplify: $\sqrt{(216)^{\frac{2}{3}} \times \frac{(25)^{\frac{1}{2}}}{(0.04)^{\frac{-1}{2}}}}$	04	OR	If $A = \{a, b, c, d\}$, $B = \{a, c, e\}$, $U = \{a, b, c, d, e, f\}$, then show that using Venn diagram $(A \cup B)' = A' \cap B'$.	2+2												
(ii)	Two boats A and B are 5 km apart and the bearing of boat B from boat A is 250° . Draw the diagram showing relative positions of both boats, then find the bearing of boat A from boat B.	04	OR	Let $A = \{1, 4, 8\}$, $B = \{1, 0\}$, find Cartesian product $A \times B$. Find $R_1 = \{(x, y) x \in A, y \in B \wedge x \geq y\}$	3+1												
(iii)	The given data shows the distribution of weights (in kg) of 30 bags of rice. Calculate mean weight of the rice bags. <table border="1"> <tr> <td>C - I</td><td>05-09</td><td>10-14</td><td>15-19</td><td>20-24</td><td>25-29</td></tr> <tr> <td>f</td><td>5</td><td>7</td><td>10</td><td>4</td><td>4</td></tr> </table>	C - I	05-09	10-14	15-19	20-24	25-29	f	5	7	10	4	4	04	OR	Find equation of line passing through point of intersection of lines $2x + y + 2 = 0$, $x - 2y + 1 = 0$ and Parallel to $2x - 2y + 11 = 0$.	04
C - I	05-09	10-14	15-19	20-24	25-29												
f	5	7	10	4	4												
(iv)	Solve the inequality: $2x - 3 < x + 1$ and $3x - 3 \geq 2x + 8$	04	OR	Factorize: $(x^2 - 7x + 4)(x^2 - 7x + 5) - 2$	04												
(v)	Show that: $1 + \tan^2 \theta = \sec^2 \theta$	04	OR	Find the gradient of the line passing through $(-4, -3)$ and $(2, 9)$. Also write equation of this line.	04												
(vi)	In a certain polygon, the sum of the measures of all the interior angles is equal to twice that of the exterior angles. What is the name of that polygon? If one side is 1 m then find this perimeter.	2+1 +1	OR	A number is chosen at random from a set of whole numbers from 1 to 50. Calculate the probability: a. Chosen number is not a perfect square. b. Chosen number is perfect square.	2+2												
(vii)	A water fountain is to be placed at a point which is equidistant from two lamp posts located at $P(3, 7)$ and $Q(9, 1)$. Find possible positions of the fountain.	3+1	OR	Find the equation of straight-line having y-intercept = 4 and parallel to the line with slope $\frac{1}{2}$.	04												
(viii)	In the figure two geometrically similar cylinders are shown. Find: 	2+1 +1	OR	Starting with 500 users, a startup experiences continuous growth at a rate of 8% per year. To determine the number of years it takes for the company to reach 3000 users, use the formula $N(t) = N_0 \times e^{rt}$ where $N(t)$ is the number of users after time t . N_0 is the initial number of users, r is the growth rate and t is the time in years.	04												
(ix)	If $A = 4a^2b$ and $B = 6ab^2$ then verify that $A \times B = LCM \times HCF$	2+2	OR	The populations (in thousands) of nine towns are: 24, 18, 30, 22, 26, 26, 26, 32, 36. Find Mode Median Mean	1+1 +2												

SECTION – C (Marks 24)

Note: Attempt the following questions.

(3x8=24)

Q.3	To convert $x^4 + 4x^3 + 10x^2 + 10x + 5$ into a perfect square: a. What should be added to it? b. What should be subtracted from it?	08	OR	Find the interior angles of triangle XYZ whose vertices are: $X(-2, 0)$, $Y(1, -4)$, $Z(6, 6)$	08												
Q.4	Two men on opposite sides of a tree are in line with it. They observe that angles of elevation of top of tree are 30° and 60° . The height of tree is 15 m, find distance between the men.	08	OR	Convert the $3x + 4y - 12 = 0$ in given forms: a. Slope intercept b. Two intercepts c. Symmetric d. Normal	2x4												
Q.5	Draw a triangle with lengths, 2cm, 3cm, 4cm. a. Draw perpendicular bisectors. b. Are these bisectors concurrent? Verify.	3+3 +2	OR	The daily consumption of electricity units of 35 factory machines is recorded as in given table. Calculate mode of the data. <table border="1"> <tr> <td>C - I</td><td>20-24</td><td>25-29</td><td>30-34</td><td>35-39</td><td>40-44</td></tr> <tr> <td>f</td><td>8</td><td>12</td><td>2</td><td>10</td><td>3</td></tr> </table>	C - I	20-24	25-29	30-34	35-39	40-44	f	8	12	2	10	3	08
C - I	20-24	25-29	30-34	35-39	40-44												
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